Matias Pautasso

Full Stack Developer - Backend & Distributed Systems

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Professional Summary

Full Stack Developer specialized in backend engineering, distributed systems and business software. I build reliable, well-tested services in TypeScript with NestJS and PostgreSQL, applying message queues, microservices and consistency patterns (transactional outbox, orchestrated saga, idempotency). Experienced in translating real business processes - inventory, sales, cash and treasury - into production software. Strong foundations in software architecture, data modeling and automated testing.

Technical Skills

Languages: TypeScript, JavaScript, SQL, PL/pgSQL

Backend: Node.js, NestJS, REST APIs, gRPC, Protocol Buffers, WebSockets, Microservices

Frontend: React, Next.js, Vite, HTML, CSS

Databases: PostgreSQL, Prisma ORM, Redis, pgvector, database design

Messaging & async: BullMQ, Redis queues, transactional outbox, saga, idempotency

AI: RAG, vector search, AI agents & tool calling, Model Context Protocol (MCP), Vercel AI SDK

DevOps & tooling: Docker, Docker Compose, GitHub Actions, CI/CD, Git, monorepos

Testing: Jest, Supertest, Playwright, Testcontainers, integration & end-to-end testing

Architecture: Hexagonal architecture, domain modeling, clean/screaming architecture, TDD

Selected Projects

Business ERP - Los Hermanos (Client project)

TypeScript, NestJS, PostgreSQL, Prisma, React, Playwright, Docker

- Production ERP for a real retail business covering inventory, point-of-sale, and cash & treasury control.
- Built as a NestJS + PostgreSQL monorepo with a responsive web client and a Windows desktop distribution.
- Implemented Excel import/export, weight-based selling and fast point-of-sale flows aligned to daily operations.
- Covered by unit, integration and end-to-end (Playwright) tests; deep business-process and product knowledge.

JobFlow - Distributed Job Processing Platform

NestJS, BullMQ, Redis, PostgreSQL, WebSockets, Turborepo

- Asynchronous bulk CSV import processed in the background by an independent NestJS worker.
- Transactional outbox for reliable publishing; idempotent, resumable imports with checkpoints.
- Automatic retries with exponential backoff and real-time progress streamed over WebSockets.

Distributed Orders System

NestJS, gRPC, Protocol Buffers, PostgreSQL, BullMQ, Docker

- Order-processing backend as independent microservices with a database per service.
- Synchronous gRPC with Protobuf; orchestrated saga with reconciliation and transactional outbox.
- Atomic stock reservation with row locks - no overselling under concurrency; idempotent requests.

AI Business Assistant

NestJS, React, pgvector, Vercel AI SDK, MCP, Prisma

- Enterprise-style assistant answering over internal docs and business data via a RAG pipeline (pgvector).
- Multi-step AI agent with tool calling; Model Context Protocol server reusing the same domain logic.
- Hexagonal, fully typed and fully tested TypeScript monorepo with an evaluation suite.

AI Sales Assistant

Next.js, Vercel AI SDK, OpenAI, pgvector, Prisma, shadcn/ui

- B2B sales assistant handling product discovery, pricing and lead capture in a single streaming chat.
- Multi-step tool calling and semantic product search with pgvector; lead export to Google Sheets.

Education

Systems Analysis (Analista de Sistemas), in progress - Cordoba, Argentina.

Languages

Spanish (native), English (professional working proficiency).

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Desarrollador Full Stack - Backend y Sistemas Distribuidos

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Perfil Profesional

Desarrollador Full Stack especializado en ingenieria backend, sistemas distribuidos y software de negocio. Construyo servicios confiables y testeados en TypeScript con NestJS y PostgreSQL, aplicando colas de mensajes, microservicios y patrones de consistencia (transactional outbox, saga orquestada, idempotencia). Con experiencia traduciendo procesos de negocio reales - inventario, ventas, caja y tesoreria - en software de produccion. Bases solidas en arquitectura de software, modelado de datos y testing automatizado.

Habilidades Tecnicas

Lenguajes: TypeScript, JavaScript, SQL, PL/pgSQL

Backend: Node.js, NestJS, APIs REST, gRPC, Protocol Buffers, WebSockets, Microservicios

Frontend: React, Next.js, Vite, HTML, CSS

Bases de datos: PostgreSQL, Prisma ORM, Redis, pgvector, diseno de bases de datos

Mensajeria y async: BullMQ, colas Redis, transactional outbox, saga, idempotencia

IA: RAG, busqueda vectorial, agentes de IA y tool calling, Model Context Protocol (MCP), Vercel AI SDK

DevOps y tooling: Docker, Docker Compose, GitHub Actions, CI/CD, Git, monorepos

Testing: Jest, Supertest, Playwright, Testcontainers, testing de integracion y end-to-end

Arquitectura: Arquitectura hexagonal, modelado de dominio, clean/screaming architecture, TDD

Proyectos Destacados

ERP de Negocio - Los Hermanos (Proyecto de cliente)

TypeScript, NestJS, PostgreSQL, Prisma, React, Playwright, Docker

- ERP en produccion para un comercio real: inventario, punto de venta y control de caja y tesoreria.
- Monorepo NestJS + PostgreSQL con cliente web responsive y distribucion de escritorio para Windows.
- Import/export de Excel, venta por peso y flujos rapidos de punto de venta alineados a la operacion diaria.
- Cubierto por tests unitarios, de integracion y end-to-end (Playwright); fuerte conocimiento de negocio y producto.

JobFlow - Plataforma de Procesamiento Distribuido de Jobs

NestJS, BullMQ, Redis, PostgreSQL, WebSockets, Turborepo

- Importacion masiva de CSV asincrona procesada en segundo plano por un worker NestJS independiente.
- Transactional outbox para publicacion confiable; importaciones idempotentes y reanudables con checkpoints.
- Reintentos automaticos con backoff exponencial y progreso en tiempo real por WebSockets.

Sistema de Ordenes Distribuido

NestJS, gRPC, Protocol Buffers, PostgreSQL, BullMQ, Docker

- Backend de procesamiento de ordenes como microservicios independientes con una base por servicio.
- gRPC sincrono con Protobuf; saga orquestada con reconciliacion y transactional outbox.
- Reserva de stock atomica con row locks - sin sobreventa bajo concurrencia; requests idempotentes.

Asistente de Negocio con IA

NestJS, React, pgvector, Vercel AI SDK, MCP, Prisma

- Asistente de estilo empresarial que responde sobre documentacion interna y datos de negocio con RAG (pgvector).
- Agente de IA multi-paso con tool calling; servidor Model Context Protocol que reutiliza la misma logica de dominio.
- Monorepo TypeScript hexagonal, totalmente tipado y testeado, con suite de evaluacion.

Asistente de Ventas con IA

Next.js, Vercel AI SDK, OpenAI, pgvector, Prisma, shadcn/ui

- Asistente de ventas B2B que resuelve descubrimiento de productos, precios y captura de leads en un chat en streaming.
- Tool calling multi-paso y busqueda semantica de productos con pgvector; exportacion de leads a Google Sheets.

Educacion

Analista de Sistemas, en curso - Cordoba, Argentina.

Idiomas

Espanol (nativo), Ingles (competencia profesional de trabajo).